

ECA1411-EMBEDDED SYSTEM

LAB MANUAL

Exp No:27- BINARY SEMAPHORE IN SYSTEMC

```
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Hit:2 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.6 kB]
Get:5 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.2 kB]
Get:6 http://archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [97.9 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [188 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Components [1056 B]
Fetched 788 kB in 5s (150 kB/s)
118 packages can be upgraded. Run 'apt list --upgradable' to see them.
samrakshini_g@LAPTOP-K4N9FU0J:~/systemc_test$ mkdir -p ~/binary_semaphore
samrakshini_g@LAPTOP-K4N9FU0J:~/systemc_test$ cd ~/binary_semaphore
samrakshini_g@LAPTOP-K4N9FU0J:~/binary_semaphore$ nano binary_semaphore.cpp
samrakshini_g@LAPTOP-K4N9FU0J:~/binary_semaphore$ g++ -o binary_semaphore binary_semaphore.cpp -lsystemc
samrakshini_g@LAPTOP-K4N9FU0J:~/binary_semaphore$ ls
binary_semaphore  binary_semaphore.cpp
samrakshini_g@LAPTOP-K4N9FU0J:~/binary_semaphore$ ./binary_semaphore

SystemC 3.0.2-Accellera --- Nov  5 2025 12:44:46
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Task1 Entered Critical Section at 1 s
Task1 Entered Critical Section at 4 s
Task1 Entered Critical Section at 7 s
samrakshini_g@LAPTOP-K4N9FU0J:~/binary_semaphore$
```